

Basin MCMC Assignment 1

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Due on Thursday, 10 April 2025 10AM

The purpose of this assignment is to break down the development of the initial basin probability distribution and probability in the Basin MCMC algorithm.

Question 1.**[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top$. Provide a function that approximates the log posterior $\log p(\theta | y)$ on $\mathcal{X}_1$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_1 : \mathcal{X}_1 \rightarrow \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. a good approximation to the log posterior in the region of high probability around θ_0 ;
3. easy to compute defining quantities from θ_0 ;
4. easy to sample from;
5. as simple as possible.

Solution. **TODO: read over.**

Question 2.**[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top$. Provide a function that approximates the log likelihood $\log p(y | \theta)$ on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $g_2 : \mathcal{X}_2 \rightarrow \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable, i.e., $g_2(\theta_1, \theta_2) = g_2(\theta_2, \theta_1)$;
3. a good approximation to the log likelihood in the region of high probability around θ_0 ;
4. easy to compute defining quantities from θ_0 ;
5. easy to sample from;
6. as simple as possible.

Solution. **TODO: read over.**

Question 3.**[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top$. Provide a function that approximates the log posterior $\log p(\theta | y)$ on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_2 : \mathcal{X}_2 \rightarrow \mathbb{R}$ should satisfy the following properties:

1. unimodal;

2. exchangeable, i.e., $f_2(\theta_1, \theta_2) = f_2(\theta_2, \theta_1)$;
3. a good approximation to the log posterior in the region of high probability around θ_0 ;
4. easy to compute defining quantities from θ_0 ;
5. easy to sample from;
6. as simple as possible.

This should build on the function found in Question 2.

Solution.

Question 4.

[pts.]

Fix $n \geq 2$. Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top$. Provide a function that approximates the log posterior $\log p(\theta | y)$ on $\mathcal{X}_n$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_n : \mathcal{X}_n \rightarrow \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable;
3. a good approximation to the log posterior in the region of high probability around θ_0 ;
4. easy to compute defining quantities from θ_0 ;
5. easy to sample from;
6. as simple as possible.

This should be a generalization of the function found in Question 3.

these should be proportional to posterior up to a multiplicative constant?

Solution.

Question 5.

[pts.]

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top$. Rewrite the function found in Question 1 in the same form as the functions found in Question 4.

If it is not possible to write the function found in Question 1 in the same form as the functions found in Question 4, explain why. Then, provide an alternative function that approximates the log posterior $\log p(\theta | y)$ on $\mathcal{X}_1$ that is in the same form as the functions found in Question 4.

Solution.

Question 6.

[pts.]

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top$. Provide an approximation of the sequence of probabilities $\{p_n\}_{n \in \mathbb{N}}$ where

$$p_n := P(\theta \in X_n | y) = \int_{\mathcal{X}_n} p(\theta | y) d\theta,$$

where $y = A(\theta_0)$ is the noise-free data generated by θ_0 .

You may find it useful to use the functions found in Questions 4 and 5. Note that you need only approximate the sequence up to a common multiplicative constant; that is, it suffices to approximate quantities proportional to $\{p_n\}_{n \in \mathbb{N}}$.

Solution.

Question 7.**[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top$. Provide an approximation of the probability p where

$$p :=,$$

where $y = A(\theta_0)$ is the noise-free data generated by θ_0 .

Solution.